8.8.3 Isobaric process

This is a thermodynamic process that occurs at constant pressure. Even though pressure is constant in this process, temperature, volume and internal energy are not constant. From the ideal gas equation, we have

$$V = \left(\frac{\mu R}{P}\right) T \qquad (8.43)$$

Here $\frac{\mu R}{P} = constant$

In an isobaric process the temperature is directly proportional to volume.

$$V \propto T \quad \text{(Isobaric process)} \qquad (8.44)$$

This implies that for a isobaric process, the V-T graph is a straight line passing through the origin.

If a gas goes from a state (V_i, T_i) to (V_f, T_f) at constant pressure, then the system satisfies the following equation

$$\frac{T_f}{V_f} = \frac{T_i}{V_i} \qquad (8.45)$$

Examples for Isobaric process:

(i) When the gas is heated and pushes the piston so that it exerts a force equivalent to atmospheric pressure plus the force due to gravity then this process is isobaric. This is shown in Figure 8.33

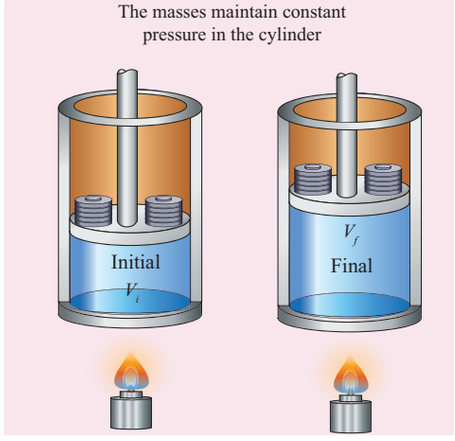


Figure 8.33 Isobaric process

(ii) Most of the cooking processes in our kitchen are isobaric processes. When the food is cooked in an open vessel, the pressure above the food is always at atmospheric pressure. This is shown in Figure 8.34



Figure 8.34 Isobaric process

The PV diagram for an isobaric process is a horizontal line parallel to volume axis as shown in Figure 8.35.

Figure 8.35 (a) represents isobaric process where volume decreases

Figure 8.35 (b) represents isobaric process where volume increases

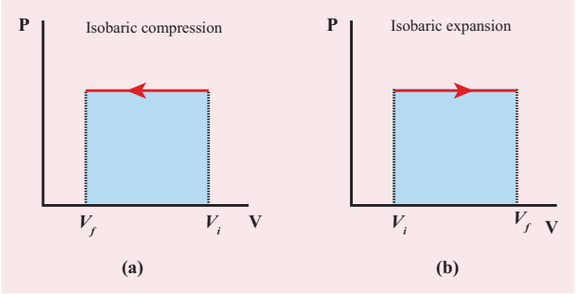


Figure 8.35 PV diagram for an isobaric process

The work done in an isobaric process:

Work done by the gas

$$W = \int_{V_i}^{V_f} P dV \qquad (8.46)$$

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